

Data Flow Diagram & User Stories

Date	29 September 2026
Team ID	SWTID-2026-8776
Project Name	AI WeatherWise API
Maximum Marks	4 Marks

Chevalier T. Thomas Elizabeth College for Women

Team:

Sairunisha S M – Team Leader

Swetha R

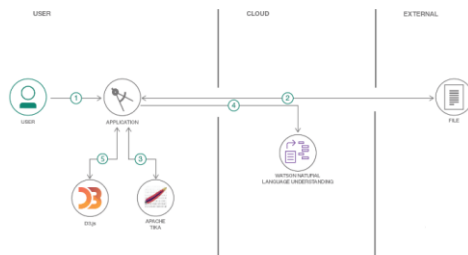
Lokeshwari P

Yuvas shri R

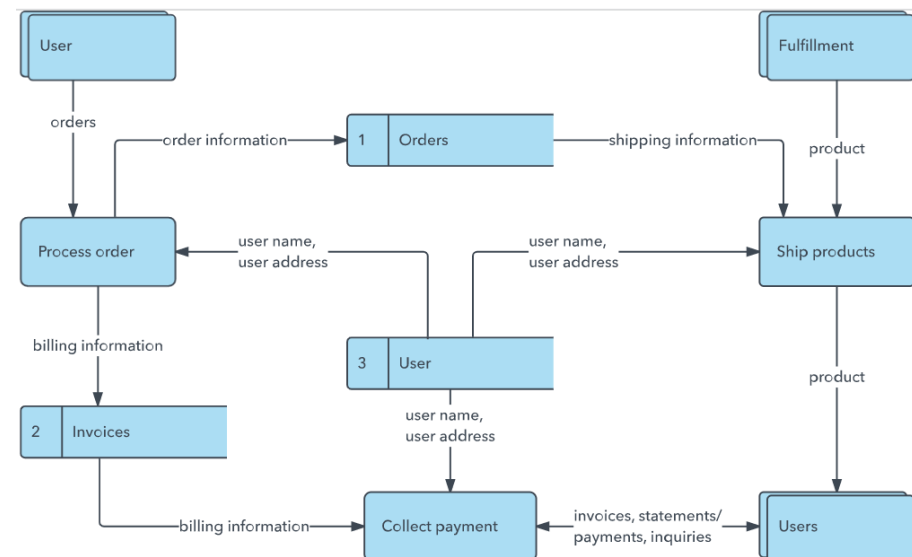
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer / API User	User Registration	USN-1	As a user, I can register by providing my name, email and password.	Valid details create a new user account and store the password securely.	High	Sprint-1
Customer / API User	Authentication	USN-2	As a user, I can log in using my registered email and password.	Valid credentials generate a JWT token and allow access to protected APIs.	High	Sprint-1
Customer / API User	Authentication	USN-3	As a user, I can access protected services using my JWT token.	Requests without a valid token are rejected; valid tokens allow access.	High	Sprint-1
Customer / API User	User Profile	USN-4	As a user, I can view my profile information.	The API returns the authenticated user's profile information.	Medium	Sprint-2
Customer / API User	Weather Information	USN-5	As a user, I can retrieve real-time weather information for a city.	The API returns available weather data such as temperature, humidity, wind and conditions.	High	Sprint-3
Customer / API User	AI Weather Summary	USN-6	As a user, I can receive an AI-generated summary of the current weather.	Gemini processes the weather information and returns an understandable summary.	High	Sprint-4
Customer / API User	AI Recommendations	USN-7	As a user, I can receive personalized recommendations based on weather conditions.	The API provides relevant suggestions for clothing, activities, travel and safety.	High	Sprint-4
Customer / API User	Favorite Locations	USN-8	As a user, I can save a city as a favorite location.	The location is stored against the authenticated user's account.	High	Sprint-2
Customer / API User	Favorite Locations	USN-9	As a user, I can view my saved favorite locations.	The API returns the locations saved by the authenticated user.	High	Sprint-2
Customer / API User	Favorite Locations	USN-10	As a user, I can update a saved location.	The selected location details are updated successfully.	Medium	Sprint-2
Customer / API User	Favorite Locations	USN-11	As a user, I can delete a saved location.	The selected location is removed from the user's saved locations.	Medium	Sprint-2